Customer Segmentation & Retention Strategy (RFM Analysis)

1. Business Problem:

The objective of this analysis is to understand customer purchase behaviour using RFM (Recency, Frequency, Monetary) analysis and identify high-value, loyal, and at-risk customers. The goal is to help the business improve customer retention, increase repeat purchases, and maximize long-term revenue through data-driven customer segmentation.

2. RFM Analysis:

Recency:

Measures how recently a customer made a purchase.

Formula-

Recency = Analysis Date - Last Purchase Date

Lower recency indicates more active customer

Frequency:

Measures how often a customer purchases

Formula-

Frequency = Total Completed Orders

Higher frequency indicates stronger customer engagement.

Monetary:

Measures total revenue generated by a customer.

Formula-

Revenue = Quantity × Unit Price × (1 − Discount)

Higher monetary value indicates high-value customers.

3. RFM Scoring:

Each RFM metric was converted into scores from 1 to 5 using quantile-based scoring (qcut).

Scoring Logic

- Recency Score
  Lower recency → Higher score
  More recent customers received score 5
  Less recent customers received score 1
- Frequency Score
  Higher purchase frequency → Higher score
- Monetary Score
  Higher spending → Higher score

4. Customer Segmentation:

Customers were segmented using RFM scores.

Segmentation Logic

- ***Champions***
  Customers with: High recency, High frequency and High monetary value.
  These are the most valuable customers.

- ***Loyal Customers***
  Customers with: Good recency, Consistent purchasing behaviour and Moderate to high spending.
  These customers contribute stable long-term revenue.

- ***Potential Loyalists***
  Customers with: Recent purchases and Lower frequency.
  These customers have potential to become loyal customers.

- ***At Risk***
  Customers with: Low recency, Previously high frequency and monetary value.
  These customers are likely to churn.

- ***Lost Customers***

Customers with: Low recency, Low frequency and Low monetary value.
These customers show minimal engagement.

5. Business Analysis:

<u>Segment Performance Summary</u>

- **At Risk**
  Customer Count: 98
  Customer Share: 23.90%
  Total Revenue: ₹1,598,104
  Revenue Share: 28.37%
  Average Order Value: ₹8,153.59
  Purchase Frequency: 2.00

- **Champions**
  Customer Count: 61
  Customer Share: 14.88%
  Total Revenue: ₹1,519,941
  Revenue Share: 26.98%
  Average Order Value: ₹7,272.45
  Purchase Frequency: 3.43

- **Lost Customers**
  Customer Count: 118
  Customer Share: 28.78%
  Total Revenue: ₹565,520
  Revenue Share: 10.04%
  Average Order Value: ₹3,448.29
  Purchase Frequency: 1.39

- **Loyal Customers**
  Customer Count: 85

Customer Share: 20.73%
Total Revenue: ₹1,616,756
Revenue Share: 28.70%
Average Order Value: ₹7,348.89
Purchase Frequency: 2.59

- **Potential Loyalists**
  Customer Count: 48
  Customer Share: 11.71%
  Total Revenue: ₹332,324
  Revenue Share: 5.90%
  Average Order Value: ₹5,729.73
  Purchase Frequency: 1.21

Segment Priority Ranking

1. Loyal Customers:
Loyal Customers generate the highest revenue contribution (28.70%) and show strong purchase frequency. This segment should be prioritized to maximize long-term stable revenue.

2. At Risk Customers:
At Risk customers contribute 28.37% of revenue. Retaining this segment is critical because losing them can directly impact business performance.

3. Champions:
Champions have the highest purchase frequency (3.43 purchases/customer) and contribute 26.98% of total revenue. These are highly valuable repeat customers.

4. Potential Loyalists:
Potential Loyalists show recent engagement but lower purchase frequency. This segment has strong future growth potential.

5. Lost Customers:
Lost Customers represent the largest customer group (28.78%) but contribute only 10.04% revenue, making them the lowest business priority

6. Key Insights:
Insight 1
Observation → Lost Customers are the largest customer segment.
Data → Lost Customers represent 28.78% of total customers (118 customers).
Business Impact → A large portion of customers show weak engagement, increasing customer churn risk.

Insight 2
Observation → Champions contribute high revenue despite a smaller customer base.
Data → Champions represent only 14.88% of customers (61 customers) but contribute 26.98% of total revenue (₹1,519,941).
Business Impact → Revenue is highly dependent on high-value customers, making retention extremely important.

Insight 3
Observation → At Risk customers generate significant revenue.
Data → At Risk customers contribute 28.37% of total revenue (₹1,598,104).
Business Impact → Losing this segment can create major revenue loss for the business.

Insight 4
Observation → Champions have the highest purchase frequency.
Data → Champions average 3.43 purchases per customer.

Business Impact → Highly engaged customers improve repeat purchases and customer lifetime value.

Insight 5
Observation → Lost Customers have the lowest average order value.
Data → Lost Customers generate only ₹3,448 average order value.
Business Impact → Low-value customers contribute limited profitability and weak long-term business growth.

Insight 6
Observation → Potential Loyalists have recent activity but low purchasing frequency.
Data → Potential Loyalists represent 11.71% of customers with only 1.21 purchases per customer.
Business Impact → This segment has strong future growth potential if targeted correctly.

7. Recommendations:
Recommendation 1
Personalized VIP loyalty campaigns for Champions customers to improve repeat purchases and customer retention.
Expected Impact: Increase customer loyalty and maximize customer lifetime value.
Success Metric: Repeat Purchase Rate

Recommendation 2
Exclusive discounts and reward programs for Loyal Customers to encourage higher purchase frequency.
Expected Impact: Increase long-term stable revenue.
Success Metric: Customer Lifetime Value.

Recommendation 3

Targeted email marketing and personalized product recommendations for Potential Loyalists.
Expected Impact: Convert more customers into loyal repeat buyers.
Success Metric: Repeat Conversion Rate.

Recommendation 4
Win-back campaigns and special offers for At Risk customers.
Expected Impact: Reduce customer churn and recover declining customers.
Success Metric: Customer Retention Rate.

8. Conclusion:
The RFM analysis successfully identified valuable customer segments based on purchasing behaviour. Champions and Loyal Customers contribute the highest business value, while At Risk and Lost Customers highlight retention challenges.

The analysis shows that customer revenue is concentrated among high-value segments, making retention strategies critical for long-term business growth. By implementing targeted loyalty programs, personalized marketing campaigns, and customer retention strategies, the business can improve repeat purchases and maximize overall customer lifetime value.

A data-driven segmentation strategy enables better customer prioritization and supports smarter business decision-making.